

ITDS12053 Introduction to Data Science

Tutorial & Lesson Notes

Topic: Data Acquisition Strategy, Privacy & Data Quality

Module
ITDS12053

Faculty
Information Technology

Covers
Question 1 Exam Content

Learning Objectives

By the end of this tutorial you will be able to: (1) identify and justify data sources for specific business objectives, (2) explain the benefits of diverse data acquisition, (3) articulate data privacy principles and relevant regulations, (4) conduct due diligence on third-party data brokers, (5) perform data standardisation, (6) evaluate data quality issues and apply appropriate remediation strategies, and (7) analyse cost trade-offs in data acquisition decisions.

SECTION 1: Understanding Data Sources

1.1 What is a Data Source?

A data source is any origin from which data is collected and stored for use in analysis or business operations. In a real-world organisations, data sources range from internal operational systems to external third-party feeds.

Types of Data Sources

Term	Definition
Internal Sources	Data generated within the organisation — e.g., ERP, CRM, website analytics, IoT sensors in physical stores.
External Sources	Data acquired from outside the organisation — e.g., purchased demographic datasets, social media APIs, public government data.
Primary Data	Data collected directly by the organisation for a specific purpose — e.g., customer feedback surveys.
Secondary Data	Data collected by another party but repurposed — e.g., buying consumer behaviour data from a data broker.
Structured Data	Data organised in a predefined schema, typically stored in tables — e.g., transaction records in a database.

Term	Definition
Unstructured Data	Data without a predefined format — e.g., social media posts, images, free-text reviews.
Semi-structured Data	Data with some organisational properties — e.g., JSON files, XML logs from web analytics.
Real-time / Streaming	Data generated and transmitted continuously — e.g., IoT foot traffic sensor readings.
Batch Data	Data collected and processed at scheduled intervals — e.g., daily sales reports from point-of-sale systems.

1.2 Matching Data Sources to Business Objectives

A core skill for any data analyst is the ability to select the most relevant data sources for a specific business objective. The key is to understand what type of insight is needed and work backwards to the data that can generate it.

Framework: Business Objective → Data Needed → Best Source

Business Objective	Recommended Data Sources & Justification
Reduce e-commerce cart abandonment	Website analytics (tracks where users drop off the funnel), session recordings (reveals friction in UX), customer feedback surveys (captures reasons for abandonment in users' own words).
Improve customer satisfaction	CRM system (history of complaints and resolutions), feedback surveys (direct ratings), social media sentiment (unprompted opinions).
Optimise in-store layouts	IoT foot traffic counters (measures where customers walk), smart shelf sensors (tracks which products are picked up), POS data (sales by location in store).
Personalise product recommendations	Purchase history from ERP, browsing behaviour from website analytics, demographic data from CRM or third-party brokers.
Plan new store locations	Public demographic data (population density, income levels), competitor location data, foot traffic from existing nearby stores.

Exam Tip — Justifying a Data Source

When an exam question asks you to 'identify and justify' data sources, always structure your answer in three parts: (1) Name the source, (2) State what specific data it contains, and (3) Explain the exact insight that data will produce for the stated objective. A source name alone is worth very few marks.

1.3 Why Multiple Diverse Data Sources Are Better Than One

Relying on a single data source creates blind spots. A comprehensive business decision requires triangulating insights across multiple sources for the following reasons:

- **Corroboration:** Multiple sources allow findings to be cross-validated. If POS data shows a product is selling poorly and social media data independently shows negative sentiment about that product, the evidence is stronger than either signal alone.
- **Completeness:** No single system captures all aspects of customer behaviour. A CRM captures purchase transactions but not browsing behaviour before the purchase. Website analytics adds that missing layer.
- **Resilience:** If one data source becomes temporarily unavailable (e.g., a CRM server outage), analysis can continue using other sources.
- **Avoiding Sampling Bias:** A single survey may only reach customers who choose to respond, skewing results. Combining it with behavioural data from all users eliminates this bias.
- **Temporal Richness:** Different sources capture data at different points in the customer journey, providing a fuller timeline from awareness to purchase to post-sale feedback.

RetailConnect Example

RetailConnect wants to understand why sales dropped in Q3. Relying only on POS (point-of-sale) data shows the 'what' — units sold declined. Adding social media data reveals the 'why' — negative customer sentiment following a competitor's promotional campaign. Adding IoT foot traffic data adds another dimension — store footfall actually increased, suggesting customers visited but chose not to buy. No single source would have revealed this complete picture.

SECTION 2: Data Privacy, Ethics & Compliance

2.1 Why Data Privacy Matters in Data Acquisition

When organisations collect personal data — especially from customers — they take on legal and ethical responsibilities. Failing to manage privacy risks can result in regulatory fines, reputational damage, loss of customer trust, and civil litigation.

2.2 Key Privacy Regulations You Must Know

Regulation / Framework	Key Requirements
GDPR (General Data Protection Regulation) — EU	Requires a lawful basis for processing, explicit consent, right to erasure (right to be forgotten), data minimisation, breach notification within 72 hours. Applies to any organisation processing EU residents' data.
CCPA (California Consumer Privacy Act) — USA	Gives California residents the right to know what data is collected, right to delete, right to opt out of sale of personal data. Applies to companies exceeding revenue or data-processing thresholds.
HIPAA (Health Insurance Portability and Accountability Act) — USA	Governs health data specifically. Requires safeguards for Protected Health Information (PHI). Relevant when health data is shared across organisations.
PDPA (Personal Data Protection Act)	Adopted in various Asian countries (Singapore, Thailand, Sri Lanka). Requires consent for collection, limits purpose, and grants individuals access rights.
ISO 27001	International standard for information security management. Provides best-practice framework for protecting data assets.

2.3 Core Data Privacy Principles

Regardless of which specific regulation applies, most share a common set of foundational principles:

- **Lawfulness, Fairness and Transparency:** Data must be collected with a valid legal basis. Customers must be told how their data will be used in clear, plain language (privacy notices).
- **Purpose Limitation:** Data collected for one specific purpose (e.g., processing a purchase) must not be repurposed for a different use (e.g., selling to advertisers) without additional consent.
- **Data Minimisation:** Collect only the data that is strictly necessary for the stated purpose. Collecting extensive location histories for a newsletter subscription is disproportionate.
- **Accuracy:** Organisations must take reasonable steps to ensure personal data is accurate and kept up to date.
- **Storage Limitation:** Personal data should not be retained longer than necessary. Retention policies should define how long each data type is kept.
- **Integrity and Confidentiality:** Data must be protected against unauthorised access, loss, or destruction using appropriate technical and organisational security measures.

- **Accountability:** The organisation must be able to demonstrate compliance with all the above — not merely assert it.

2.4 Specific Challenges When Collecting Customer Data Online

The marketing team at RetailConnect wants to collect email addresses, purchase history, browsing behaviour, and location data. Each type raises distinct privacy considerations:

Data Type	Privacy Consideration
Email Addresses	Requires explicit consent (opt-in) before use for marketing. Unsubscribe mechanism must be provided. Should be encrypted at rest and in transit.
Purchase History	Is a form of sensitive behavioural data. Must be protected and only used for stated purposes. Retention period must be defined.
Browsing Behaviour	Collected via cookies or tracking pixels. Under GDPR and CCPA, consent banners are legally required. Users must be able to reject non-essential tracking.
Location Data	Considered sensitive under most regulations. Granular, real-time location data is particularly sensitive — it can reveal patterns about where a person lives, worships, or seeks medical care. Explicit consent and clear purpose are essential.

2.5 Due Diligence for Third-Party Data Brokers

When purchasing external data from a data broker, the acquiring organisation does not eliminate its legal responsibility — it shares it. Under GDPR and similar laws, if you use unlawfully obtained data, you are also in violation. The following questions must be asked of any data broker:

- **1. How was this data originally collected?** Ask for documentation of the original consent mechanisms. If the broker cannot demonstrate that individuals consented to their data being sold to third parties, the data should not be purchased.
- **2. Does the data collection comply with applicable regulations?** Specifically ask which privacy laws the broker operates under and whether they have conducted Privacy Impact Assessments (PIAs).
- **3. What personal identifiers are included?** Understand whether the dataset contains directly identifying information (names, emails, national ID numbers) or whether it has been anonymised or pseudonymised.
- **4. What are the contractual use restrictions?** Some datasets are licensed for specific purposes only. Using data beyond the agreed scope is a legal violation.
- **5. How current and accurate is the data?** Outdated demographic data can produce misleading analysis and waste resources. Ask for data age and update frequency.
- **6. Has the broker experienced any data breaches?** Purchasing data from a compromised broker is a security risk and may create regulatory liability.

Key Principle

The purchasing organisation is considered a 'data controller' and cannot outsource legal responsibility to the broker. Performing thorough due diligence is both an ethical obligation and a legal requirement under most modern data protection laws.

SECTION 3: Data Standardisation & Quality

3.1 What is Data Standardisation?

Data standardisation is the process of transforming data from different sources, formats, or systems into a consistent, unified format so that it can be compared, merged, and analysed reliably. Without standardisation, the same real-world entity (a customer, a product, a location) can appear in dozens of incompatible representations across your systems.

Analogy

Imagine measuring the length of three objects — one in inches, one in centimetres, and one in millimetres. You cannot simply add the three numbers together. You must first convert them all to the same unit. Data standardisation is the data equivalent of choosing a common unit of measurement.

3.2 Common Standardisation Problems and Solutions

Problem 1: Inconsistent Geographic Data

Inconsistency	Standardisation Solution
CRM: 'California', E-commerce: 'CA'	Choose one canonical form (recommended: ISO 3166-2 two-letter code 'CA'). Apply a lookup/mapping table to convert all full state names to codes across every data source. Document the chosen standard.
City names with typos: 'Los Angelas', 'Los Angeles'	Use fuzzy matching algorithms and canonical reference lists (e.g., USPS city name database) to correct spelling variations.
Postcodes in different formats: '90210' vs '90210-1234'	Strip the four-digit extension and standardise to 5-digit ZIP, or consistently use ZIP+4. Apply regex pattern matching to enforce the standard.

Problem 2: Inconsistent Phone Number Formats

RetailConnect's data contains: '(123) 456-7890', '123-456-7890', '1234567890'. These are three representations of the same number.

- **Step 1 — Strip all non-numeric characters:** Remove parentheses, hyphens, spaces. All three examples above reduce to '1234567890'.
- **Step 2 — Validate digit count:** A US domestic number has 10 digits. Numbers with 11 digits starting with '1' are US numbers with the country code — remove the leading '1' to produce 10 digits.

- **Step 3 — Reformat to canonical form:** Apply a consistent display format, e.g., '+1 (123) 456-7890' for international storage or '1234567890' for pure storage.
- **Step 4 — Flag anomalies:** Numbers with fewer than 10 digits or non-standard patterns should be flagged as potentially erroneous rather than discarded immediately.

Problem 3: Inconsistent Timestamps and Time Zones

IoT sensors in different store locations record timestamps in their local time zone. Without standardisation, a 'peak hour' analysis will produce meaningless results — 9 AM in New York is 6 AM in Los Angeles.

- **Step 1 — Identify the source time zone:** Document which time zone each store location uses. This metadata must come from the data providers.
- **Step 2 — Convert all timestamps to UTC:** UTC (Coordinated Universal Time) is the universal reference. A timestamp of '09:00 EST' becomes '14:00 UTC'. Store UTC as the canonical form in all systems.
- **Step 3 — Use ISO 8601 format:** The standard format is 'YYYY-MM-DDTHH:MM:SSZ' (e.g., 2024-03-15T14:00:00Z). This is unambiguous, internationally recognised, and sortable.
- **Step 4 — Reconstruct local time only at presentation:** When displaying data to a specific store manager, convert from UTC to their local time zone. Store data always in UTC.

Why This Matters

If timestamps are not standardised, any time-based analysis (peak trading hours, delay patterns, seasonal trends) will contain systematic errors. A 'midnight' anomaly in the data may simply be a time zone conversion error, not a real event — leading to incorrect business decisions.

3.3 The Consequences of Not Standardising Data

- **Incorrect aggregations:** Totals and averages will be wrong if units or formats are inconsistent.
- **Failed record linkage:** The same customer may be stored as two separate records because their address appears in two different formats, leading to duplicate communications and inaccurate customer counts.
- **Misleading visualisations:** Charts plotted from non-standardised data will show patterns that are artefacts of the inconsistency, not real business phenomena.
- **Broken machine learning models:** Models trained on inconsistent data will learn spurious patterns and fail to generalise to new data.
- **Regulatory risk:** Inconsistent records make it harder to respond accurately to Subject Access Requests under GDPR or to prove compliance during audits.

3.4 Data Quality Issues: Missing Data

Missing data is one of the most common data quality problems. However, not all missing data is equally concerning. The appropriate handling strategy depends on how much data is missing and why it is missing.

How Much Is Too Much?

% Missing	General Guidance
< 5%	Usually safe to impute (fill in estimated values) or ignore without significantly affecting results.
5% – 20%	Investigate the pattern. May be safe to impute with care, but document thoroughly.
20% – 40%	High risk. Imputation may introduce too much bias. Consider whether the variable can be used at all.
> 40%	Typically too unreliable to use in analysis. Consider excluding the variable or gathering more data.

Understanding WHY Data is Missing

- **Missing Completely At Random (MCAR):** The probability of a value being missing has no relationship to any other variable. Example: a sensor malfunctioned and randomly dropped readings. Safe to handle with imputation.
- **Missing At Random (MAR):** Missingness is related to other observed variables. Example: blood pressure is more likely to be missing for patients who visited for non-cardiovascular issues. Can often still be imputed, but the imputation model must account for the related variables.
- **Missing Not At Random (MNAR):** Missingness is related to the missing value itself. Example: patients with very high blood pressure may have refused measurement. This is the most dangerous pattern — imputation will be biased and may lead to incorrect conclusions.

Strategies for Handling Missing Data

Strategy	Description	When to Use	Risk
Listwise Deletion	Remove any record with a missing value.	When % missing is very small and MCAR.	Loss of data; introduces bias if not MCAR.
Mean/Median Imputation	Replace missing values with the column average or median.	Continuous variables, low missingness, MCAR.	Reduces variance; distorts distribution shape.
Mode Imputation	Replace missing categorical values with the most	Categorical variables, low missingness.	Can over-represent dominant categories.

Strategy	Description	When to Use	Risk
	common category.		
Multiple Imputation	Generate several plausible imputed datasets, analyse each, and pool results.	When missingness is significant and MAR.	Computationally intensive; requires expertise.
Use as Indicator Variable	Create a binary flag: 1 = was missing, 0 = was present. Use in analysis.	When missingness itself is informative (MNAR pattern).	Adds complexity to models.
Discard the Variable	If too much data is missing, exclude the variable from analysis.	When > 40% missing or MNAR with no remedy.	Loss of potentially valuable signal.

3.5 Data Quality Issues: Spam, Bots & Invalid Responses

Social media and survey data are particularly susceptible to contamination by automated bot-generated content or spam responses. If left unaddressed, these corrupt the signal in sentiment analysis and customer insight work.

How to Detect Bot/Spam Content

- Identical or near-identical responses submitted multiple times (high textual similarity).
- Responses completed in an impossibly short time (e.g., a 20-question survey completed in 8 seconds).
- Responses originating from flagged IP addresses or known spam accounts.
- Statistical patterns inconsistent with genuine human variation (e.g., all respondents give exactly the same rating to every question).
- Free-text fields containing irrelevant content, random characters, or known spam phrases.

Decision Criteria: Clean, Discard, or Use As-Is?

Decision	Criteria
Clean the data	The issue is systematic and correctable (e.g., standardisation errors, encoding problems). The original data intent is recoverable.
Discard the data	The data is irrecoverably corrupted, spam-generated, or cannot be verified as genuine. The cost of retaining it (bias risk) exceeds the cost of discarding it.
Use as-is	The quality issue is so minor (e.g., < 1% affected) that it will have negligible impact on results. Must be documented.
Flag and investigate	The cause of the issue is unclear. Flag records for domain expert review before making a final decision.

Impact on Business Decisions

Uncleaned social media data inflates apparent customer sentiment (bots often post positive or negative in extreme patterns). Survey data contaminated with incomplete responses produces means and distributions that do not reflect actual customer opinions. Decisions built on this data — such as which products to feature or which services to improve — will be systematically wrong, potentially at significant financial cost.

SECTION 4: Data Acquisition Cost-Benefit Analysis

4.1 Why Data Acquisition Has Costs

Data acquisition is not free. Even when data appears 'available', there are costs associated with collecting, transmitting, storing, processing, securing, and maintaining it. Finance departments will typically challenge large data acquisition budgets, and data scientists must be able to articulate the business value clearly.

4.2 Direct vs. Indirect Costs

Cost Category	Examples
Hardware	IoT sensors, server infrastructure, network equipment.
Software Licenses	Data management platforms, analytics tools, ETL (Extract, Transform, Load) pipelines.
Data Transmission	Network bandwidth costs for streaming real-time IoT data.
Cloud Storage	Object storage (AWS S3, Azure Blob) for raw data; database services for structured data.
Processing	Compute costs for transforming, cleaning, and querying large volumes of data.
Personnel	Data engineers to build pipelines; data analysts to maintain dashboards; data governance officers to ensure compliance.
Security & Compliance	Encryption tools, audit logging, privacy impact assessments, legal counsel.
Training	Upskilling staff to use new data tools and follow data governance policies.

4.3 Pilot Approach vs. Full Deployment: Trade-Off Analysis

When proposing large-scale data acquisition (e.g., IoT sensors across 50 stores), a phased approach starting with a pilot is a common risk management strategy. However, it also has limitations.

Factor	Full Rollout (50 stores, \$120K/yr)	Pilot (5 stores, \$15K/yr)
Data Representativeness	High — data reflects full geographic, demographic, and format diversity of the store network.	Low — 5 stores may not reflect the variety of the full estate. Results may not generalise.
Decision Quality	Decisions on store-level strategy are fully data-informed for all locations.	Decisions for the 45 non-pilot stores must still rely on intuition or manual

Factor	Full Rollout (50 stores, \$120K/yr)	Pilot (5 stores, \$15K/yr)
		reports.
Financial Risk	High upfront commitment; if technology underperforms, sunk cost is large.	Low financial exposure. Failed technology has minimal financial impact.
Scalability	Infrastructure design must handle full scale from day one.	Pilot learnings can inform a better, cheaper full-scale design.
Speed to Insight	Full network insights available immediately.	Full insights delayed until rollout is complete. Competitive disadvantage possible.
Organisational Learning	Operational processes for data at scale must be established immediately.	Gives team time to build capability and governance before full scale.

Recommended Approach

A risk-managed hybrid is often best: implement the pilot in 5 stores that are representative of the different store types (large/small, urban/suburban, high/low footfall), not just the easiest 5. Define measurable success criteria for the pilot upfront (e.g., 'If the pilot demonstrates X% improvement in inventory management within 6 months, full rollout is approved'). This satisfies the finance team's risk concerns while creating a clear path to full scale.

4.4 Additional Infrastructure and Operational Costs

Beyond the direct data acquisition cost, organisations must also budget for:

- **Data Storage and Management Infrastructure:** Raw data collected from IoT sensors, CRM systems, and web analytics must be stored in a structured, accessible way. This requires a data warehouse (e.g., Google BigQuery, Amazon Redshift) or a data lake for unstructured data. Ongoing costs include storage fees, query processing, and backup/disaster recovery.
- **Data Pipeline and ETL (Extract, Transform, Load) Infrastructure:** Data from multiple sources arrives in different formats and schedules. ETL pipelines clean, transform, and load data into the central repository automatically. Building and maintaining these pipelines requires engineering resources and platform costs (e.g., Apache Kafka for real-time streaming, Apache Airflow for batch orchestration).
- **Security and Compliance Costs:** Encryption, access controls, audit trails, and regular compliance reviews are ongoing costs that are often underestimated. A data breach can cost far more than all of these preventive measures combined.
- **Personnel and Training:** Data systems require skilled people to build, maintain, and use them. Hiring data engineers, data analysts, and a data governance officer represents a significant recurring cost.